

MemoKi: Phase 2 Report (Cues, Tree, Cue-Guided Search)

October 2026. Covers steps 2.0 to 2.4, plus step 3.0 (built, not yet run). Each section has a short Hindi explanation (हिंदी में).

How to read the status labels

- **Confirmed** = you ran it on your machine and saw the output.
- **Built, tested in my environment** = code and tests pass on my side only.
- **Not yet run / not started**.

All results are on the **validation split** (100 queries, one seed). They are development results, not final claims.

1. What Phase 2 was for

Build the actual contribution: cheap **cues** (names, dates), a **tree** that stores conversations hierarchically, and a **search** that uses the cues to skip branches before doing expensive vector comparisons, with zero LLM calls.

हिंदी में: Phase 2 प्रोजेक्ट का असली हिस्सा है। यहाँ हमने नाम और तारीख जैसे आसान "संकेत" निकाले, बातचीत को tree में रखा, और खोज ऐसी बनाई जो संकेतों से बेकार शाखाएँ पहले ही हटा दे, बिना किसी LLM के।

2. Step-by-step: what was done and what we found

Step	What was done	Result	Status
2.0 Cue coverage	Counted how many of 470 queries contain a name or a time phrase	Named entity 36%, time expression 21%, any cue 54%. 46% have no cue	Confirmed
2.0b Entity quality	Checked if the right conversation contains the query's name (100 queries, 400 distractors)	Recall 95% (37 of 39), selectivity mean 1.2%, median 0.2% : one name check removes about 99% of distractors	Confirmed (on test queries; rerun on validation still pending)
2.1 Entity extraction	spaCy extractor, name normalising, resumable disk cache	Entities cached for all 19,195 sessions , mean 21.3 per session. About 40 min per 10,000 with 3 processes	Confirmed
2.2 Time cue	Rule-based parser: "last month", "in March", "two weeks ago" become date windows	Applies to only 8% of queries (11 of 141). With 7 days slack: recall 100% (11/11), 13.7% of the pool inside the window	Confirmed
2.3 Tree, no cues	Balanced tree, 2-means splits, beam search, honest cost counting	Structure sweep, then frozen at 32 children, min fill 0.35 (height 3 at 10,000)	Confirmed

2.4 Cue-guided search	Entity and date summaries on every node, cue gate before vector comparison, flat fallback	Main results in section 4	Confirmed
3.0 Opponents	Flat search with the same cue filter, FAISS flat and HNSW, paired test	Built; 52 tests pass in my environment	Not yet run

हिंदी में: हमने पहले देखा कि कितने सवालों में संकेत हैं (54%)। फिर परखा कि नाम वाला संकेत भरोसेमंद है (95% सही, 99% बेकार चीज़ें हटती हैं)। तारीख वाला संकेत सही है पर सिर्फ़ 8% सवालों पर लागू होता है। फिर tree बनाया और उसमें संकेत जोड़े।

3. Design decisions, and why

Decision	Reason
Cues are gates with a safety valve , not hard filters	46% of queries have no cue. If no child matches at a level, pruning is skipped for that level
Every cue must pass recall and selectivity tests first	Fair ablation; each cue earns its place
Never use the dataset's <code>question_type</code> as a cue	A real user doesn't provide it. That would be data leakage
Tree is balanced (all leaves at the same depth), grows only when the root splits	Keeps depth small and the scaling claim honest
Tree structure chosen by a rule written before the results	Avoids bending the choice to fit 100 noisy queries
Count cue checks and embedding comparisons separately ; report query-prep time separately	Otherwise the method would look cheaper than it is
Cue code separate from tree code, with switches <code>use_names</code> , <code>use_time</code> , <code>use_fallback</code>	Ablations (Gap 4) become a loop, not a rewrite
Tune on validation only ; test split stays untouched	No tuning on the answer sheet

हिंदी में: हर फ़ैसले के पीछे वजह है। संकेतों को सख्त फ़िल्टर नहीं बनाया क्योंकि लगभग आधे सवालों में संकेत है ही नहीं। tree की बनावट पहले से लिखे नियम से चुनी, ताकि नतीजे देखकर नियम न बदला जाए। सारा tuning validation पर हुआ, test को छुआ तक नहीं।

4. Main results (validation, 100 queries, seed 11)

4.1 Recall@5 and cost at each store size

Size	Brute	Tree, no cues	Cue tree	Cue tree	Share of brute-
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	force	(beam 10)	(beam 10)	comparisons	force cost
1,000	0.630	0.600	0.680	162	16%
5,000	0.460	0.390	0.470	303	6.1%
10,000	0.360	0.330	0.420	308	3.1%

At beam 3 the cue tree uses 76 / 105 / 108 comparisons and scores 0.660 / 0.390 / 0.350 (the 5,000 result is 7 points below brute force, so beam 3 is too narrow there).

4.2 Queries with a cue versus without (10,000 sessions, beam 10)

Group	Brute force	Tree, no cues	Cue tree
With a usable cue (41 queries)	0.488	0.415	0.634
Without a cue (59 queries)	0.271	0.271	0.271

Queries without cues are unchanged, as designed. This was checked at every size.

4.3 Structure choice (10,000 sessions, beam 10, share of brute-force hits kept)

Max children, min fill	Height	Kept
8, 0 (first version)	7	61%
8, 0.35	5	67%
16, 0	5	72%
16, 0.35	4	75%
32, 0.35 (chosen)	3	86%

4.4 Flat-search fallback (negative result)

Setting (10,000 sessions, beam 10)	Recall@5	Comparisons
No fallback	0.420	308
Fallback at threshold 0.30	0.400	1,008
Fallback at threshold 0.40	0.380	2,408

The fallback replaced good cue-guided answers with worse flat answers, so recall fell and cost rose. It stays implemented but **off by default**. Its real test is the Gap 2 misfiling experiment.

4.5 Time

At 10,000 sessions, beam 10: search about 0.21 ms (brute force 0.97 ms). But query preparation (spaCy) costs about 5 ms, so **end to end the cue tree is currently slower**. Counts are the primary evidence; time is secondary.

हिंदी में: 10,000 बातचीत पर cue tree ने brute force का सिर्फ 3% काम करके (308 तुलनाएँ) उससे बेहतर नतीजा दिया (0.420 बनाम 0.360)। जिन सवालों में संकेत है, वहाँ फ़ायदा बढ़ा है (0.634 बनाम 0.488)। बिना संकेत वाले सवालों पर कोई बदलाव नहीं देखा जा रहा। Fallback ने उष्णता बढ़ा दी, इसलिए इसे अभी बंद नहीं है। समय के निष्पत्ति से अभी

5. Gap-by-gap status

Gap	What was done	Evidence so far	Still needed	Fulfilled?
1 Cost vs scale	Baseline table, no-cue tree, cue tree, scaling at 3 sizes	Cue tree within 2 points of brute force at 1k, 5k, 10k beam 10, using 16%, 6%, 3% of the comparisons	Test split, 3 to 5 seeds, sizes 100 and 500, more queries, FAISS/HNSW and flat+filter results, confidence intervals, 50k to 100k stress test	Not yet. Promising
2 Misfiled conversations	Fallback built; leaf pointers kept for re-filing	Fallback is negative under natural conditions	Deliberate misfiling at 5/10/20%, a fallback that merges, hand-labelled cue accuracy on about 100 conversations	No
3 Outdated information	Time parser and timestamps on every node	Data ready: 72 knowledge-update queries	Recency rule and its experiment	No
4 Fair comparison	One config, shared counters, same data/queries/model/seeds, fixed split, pre-registered rules, ablation switches	First partial ablation: no cues vs names vs names+time	Type and emotion cues (not built), full ablation table, 3 to 5 repeats with mean and std, Mem0 and HiMem on LoCoMo	Partly (foundation only)

हिंदी में: चारों में से कोई gap अभी पूरी तरह साबित नहीं हुआ। Gap 1 में सबसे अच्छी प्रगति है और नतीजे उम्मीद जगाते हैं, पर अभी सिर्फ development data है। Gap 2 और 3 के असली experiment बाकी हैं। Gap 4 की नींव तैयार है।

6. Bugs found and fixed (process evidence)

Bug	How it was found	Effect it missed
Entity cache lost a record after a crash	Test I wrote for a simulated crash	Silent loss during the 40-minute run
"a couple of" crashed the time parser	Running on real data	Check script would fail
Query-cue cache reused values across switch settings	Failing test	Wrong ablation numbers, with no error
Search time came out negative	Reading real output	Time column invalid
spaCy model load inflated the prep time	Reading real output	Misleading timing

Lesson: **tests give safety, real runs give evidence.** Each was fixed and has a test.

हिंदी में: पाँच bugs मिले और ठीक हुए। कुछ tests ने पकड़े, कुछ असली data चलाने पर दिखे। इसीलिए दोनों ज़रूरी हैं।

7. Limitations (state these in the final report)

- 1. **100 queries, one seed.** A one-point difference is one query. Final runs need more queries, several seeds and paired tests.
- 2. **The gain on cue queries may come from the cue filter, not the tree.** A flat store with the same filter may match it. Step 3.0 tests exactly this.
- 3. **Cues apply to few queries:** about 39% have an entity, about 8% a date. The rest rely on tree routing.
- 4. **Entity recall (95%) came from only 39 queries,** and was first measured on test queries. It must be rerun on validation.
- 5. **Query preparation (about 5 ms) outweighs search time.** A cheaper entity matcher is worth trying.
- 6. **Type and emotion cues are not built.** They may be weak, and that is an acceptable finding.
- 7. **Tests:** 45 confirmed on your machine; 52 tests exist after step 3.0 (not yet confirmed there).

हिंदी में: सबसे बड़ी सावधानी: अभी सिर्फ़ 100 सवाल और एक seed है, इसलिए छोटे अंतर भरोसे लायक नहीं। यह भी हो सकता है कि फ़ायदा tree से नहीं, सिर्फ़ संकेत वाले फ़िल्टर से आ रहा हो। यही हम अगले step में जाँचेंगे।

8. What comes next

Step	Content	Why
3.0 (now)	Run <code>eval.compare_check</code> : flat + cue filter, FAISS flat/HNSW, paired test	Tells us whether the tree itself earns the credit
2.5	Type and emotion cues, tested like the others	Needed for the full ablation (Gap 4)